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Theory Assignment - 6

Instructions : Use A = 0, B = 1, ... Z = 25. Show all modular arithmetic steps and clearly state the cryptanalysis model used in each problem.

1. Caesar wants to arrange a secret meeting with Mark Antony either at the Tiber River or at the colosseum. He sends the ciphertext EVIRE. However, Antony does not know the key, so he tries all possible keys.

(a.) What are the meaningful plaintexts Antony obtains?

⟹ Cipher = EVIRE

$P = (C - K) \bmod 26$

All 26 keys possible solution.

K	Plaintext	K	Plaintext
0	EVIRE	13	RIVER
1	DUHQD	14	QHUDQ
2	CTGPC	15	PGTCP
3	BSFOB	16	OFSBO
4	ARENA	17	NERAN
5	ZQDMZ	18	MDQZM
6	YPCLY	19	LCPYL
7	XOBKX	20	KBOXK
8	WNAJW	21	JANWJ
9	VMZIV	22	IZMVI
10	ULYHU	23	HYLUH
11	TKXGT	24	GXKTG
12	SJWFS	25	FWJSF

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Two worked trials:

K = 4

E - (4-4) mod 26 = 0 → A.

V - (21-4) mod 26 = 17 → R

I - (8-4) mod 26 = 4 → E

R - (17-4) mod 26 = 13 → N

E - (4-4) mod 26 = 0 → A ∴ ARENA

K = 13

E = (4-13) mod 26 = 17 → R.

V = (21-13) mod 26 = 8 → I

I = (8-13) mod 26 = 21 → V

R = (17-13) mod 26 = 4 → E

E = (4-13) mod 26 = 17 → R ∴ RIVER.

Therefore, ARENA and RIVER are the only meaningful plaintexts.

(b.) Q Where should Antony meet Caesar?

⇒ RIVER = TIBER RIVER

ARENA = COLOSSEUM

Both are meaningful, so Antony cannot decide from the ciphertext alone.

(c) What interesting point about a ciphertext only attack do you learn from this exercise?

⇒ A ciphertext only attack can give more than one meaningful plaintext when the ciphertext is short. A longer ciphertext is needed for unique answer.

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Q2. Consider an Affine cipher operating modulo 26. In a chosen-plaintext attack, the plaintext HAHAHA produces the ciphertext NONONO.

(a.) Convert the plaintext and ciphertext letters into their numerical numerical values.

⟹ H = 7 N = 13

A = 0 O = 14

Affine cipher → $C = (K_1 \cdot P + K_2) \bmod 26$

$K_1 = a$ $K_2 = b$.

(b) Form the required equations modulo 26.

⟹ $13 = (a \cdot 7 + b) \bmod 26$ ⟶ (1)

$14 = (a \cdot 0 + b) \bmod 26$ ⟶ (2)

$-1 = 7a$

$a = 7^{-1}$

$a = (-7^{-1}) \bmod 26$

$a = -15 \bmod 26$

$a = 11$

(c.) Determine the values of a and b.

⟹ $14 = b \bmod 26$

$b = 14 \bmod 26$

$b = 14$

(d.) Write the encryption function $E(x) = (ax+b) \bmod 26$.

$C = 11 \times 7 + 14 \bmod 26$

$C = 13$

$C = 13$

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Q3. The ciphertext CRWWX was encrypted using an affine cipher modulo 26. The attacker know that the plaintext begins with HA.

(a) Use the known plaintext - ciphertext ~~plaintext~~ pair to determine a and b.

⟹ H, A = 7, 0 (Plaintext)

C, R = 2, 17 (Ciphertext)

$C = (a \cdot P + b) \bmod 26$

$2 = 7(a) + b \bmod 26$

$17 = 0(a) + b \bmod 26$

$-15 = 7a$

$a = \frac{-15}{7}$

$a = -15 \times 7^{-1}$

$a = -15 \times 15 \bmod 26$

$a = 9$

$\gcd(9, 26) = 1$

$b = 17 \bmod 26$

$b = 17$ $\quad \therefore a = 9$

$b = 17$.

(b.) Write encryption function.

⟹ $C = (K_1 (P + K_2)) \bmod 26$

$C = (9 \cdot P + 17) \bmod 26$.

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(c) Find the multiplicative inverse of a modulo 26 and write the decryption function.

$\Rightarrow$ $P = K_1^{-1} \cdot (c - K_2) \bmod 26$

For W (22)

$P = K_1^{-1} (C - K_2) \bmod 26$

$P = 9^{-1} \cdot (22 - 17) \bmod 26$

$P = 3 \cdot (5) \bmod 26$

$P = 15 \Rightarrow P$

For Z (25)

$P = 9^{-1} (25 - 17) \bmod 26$

$P = 3(8) \bmod 26$

$P = 24 \Rightarrow Y$

Plaintext is HAPPY.

(d) Decrypt the complete ciphertext

$\rightarrow$ ~~Plaintext~~ CRWWZ = (2, 17, 22, 22, 25

C : $e\ 3 \cdot 2 + 1 = 7 \rightarrow H$

R : $3 \cdot 17 + 1 = 52 \equiv 0 \rightarrow A$

W : $3 \cdot 22 + 1 = 67 \equiv 15 \rightarrow P$

W : $67 \equiv 15 \rightarrow P$

Z : $3 \cdot 25 + 1 = 76 \equiv 24 \rightarrow Y$

4. Alice uses a One-Time Pad to encrypt the plaintext RIVER with the key NQXIP.

(a) Convert the plaintext and key into numerical values.

$\rightarrow$

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Plaintext RIVER : R = 17

I = 8

V = 21

E = 4

R = 17

NQXTP key : N = 13

Q = 16

X = 23

T = 19

P = 15

(b) Encrypt the message using $C_i = (P_i + K_i) \bmod 26$.

$\Rightarrow$

$R + N \Rightarrow (17 + 13) \bmod 26 = 4 \rightarrow E$

$I + Q \Rightarrow (8 + 16) \bmod 26 = 24 \rightarrow Y$

$V + X \Rightarrow (21 + 23) \bmod 26 = 18 \rightarrow S$

$E + T \Rightarrow (4 + 19) \bmod 26 = 23 \rightarrow X$

$R + P \Rightarrow (17 + 15) \bmod 26 = 6 \rightarrow G$.

Ciphertext = EYSXG

(c.) Decrypt the resulting ciphertext using:

$P_i = (C_i - K_i) \bmod 26$

$(4-13, 24-16, 18-23, 23-19, 6-15) \bmod 26 = (17, 8, 21, 4, 17)$

The value corresponds to RIVER.

(d.) Explain why the key must be random, equal in length to the plaintext and used only once.

$\Rightarrow$ Randomness: A truely random key makes ciphertext

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unpredictable. Patterns in a non random key can reveal patterns in the message.
Equal: Each plaintext character needs its own key character. A shorter key must be reused repeatedly, which can create exploitable patterns
Using only once: Reusing a key lets attacker compare ciphertexts and potentially learn relationships between plaintexts.